

Module 2

Literature Survey - Problem Definition

Course Outcome

CO 2: Conduct literature survey and define research problem

Literature survey

- In a literary survey, **students analyse critically and concisely earlier research and literature related to a particular research problem**, and utilize them for their own research purposes.
- It helps students in understanding the significance of new research and its connections to earlier work.

Literature survey

- Information gathering -
 Reading ,
 Searching ,
 Documentation

Gathering information.

- ▶ There are a number of different ways in which you can gain information and opinions from a number of different sources.
- ▶ Information can be collected from:
 - Questionnaires
 - Interviews
 - Newspapers/Magazines
 - Internet

Questionnaire

- ▶ Through using questionnaires to gather information this allows you to collect people's opinion on the topic you are researching.
- ▶ A questionnaire contains a list of questions and prompts you want to ask your participants.
- ▶ Questionnaires can bring back statistical answers so that they can be portrayed in graphs and tables.
 - Example:
 - 40% of people say they own a mobile.

Interviews

- ▶ Interviews are real life conversations between you and the respondent.
- ▶ Questions asked in an interview can be both open and closed depending on the type of answer you are looking for.
 - Open questions are used to find out more information about their opinions.
 - Closed questions are used to find out whether they agree or disagree with a answer.



Internet

- ▶ The internet has a wide range of information on it which can be very useful when researching a specific topic.
- ▶ When using information off the internet it is important to check that the website is reliable and the information your reading is relevant to your chosen topic.

Newspapers / Magazines

- ▶ Newspapers and magazines can provide you with real life issues based on your topic.
- ▶ These can be used as evidence when proving or disproving your hypothesis.

What is a journal index?

- Every author search for indexed journals to publish his/her article in, and tries to select a journal that is covered by a number of abstracting and indexing services.
- The indexation of a journal indicates its quality.
- Indexed journals are considered to be of higher scientific quality as compared to non-indexed journals.
- The indexed journals have expanded online presence, improved article discoverability, and reputable for high-quality publication in their own field.
- A journal index, also called a 'bibliographic index' or 'bibliographic database', is a list of journals organized by discipline, subject, region or other factors.
- Journal indexes can be used to search for studies and data on certain topics.
- Both scholars and the general public can search journal indexes.

Basic Indexing Standards

- All journals are checked for certain criteria before being added to an index.
- Each index has its own set of rules, but basic publishing standards include the following:
- An **International Standard Serial Number (ISSN)**. ISSNs are unique to each journal and indicate that the journal publishes issues on a recurring basis.
- An **established publishing schedule**.
- **Digital Object Identifiers (DOIs)**: DOIs are unique letter/number codes assigned to digital objects. The benefit of a DOI is that it will never change, unlike a website link.

Basic Indexing Standards

- **Copyright requirements**. A copyright policy helps to protect your work and outlines the rules for the use or sharing of your work, whether it's copyrighted or has some form of creative commons licensing.
- Other requirements can include **conflict of interest statements, ethical approval statements, an editorial board listed on the website, and published peer review policies.**

Why is Indexing Essential?

- **Accessibility**

Since the main purpose of a journal is to be accessible to a wide audience, once it is indexed by a database, it is immediately made available to all users of a database that has indexed that journal

- **Reputation**

Being accessible has a direct impact on a journal's reputation; if a journal is available for a large number of academic population, it will be considered a reliable source of high-quality information in a certain field.

- **Readership**

Every researcher will look to established, well-known databases as the first activity in his/her studies; if a journal is indexed in a known database in that researcher's field, he/she will find that easily and read it. This increases that journal's readership.

- Journal indexes are created by different organizations, such as:

Public bodies. For example, [PubMed](#) is maintained by the United States National Library of Medicine. PubMed is the largest index for biomedical publications.

Analytic companies. For example: the [Web of Science Core Collection](#) is maintained by Clarivate Analytics. The [WOS Core Collection](#) includes journals indexed in the following sub-indexes: (1) Science Citation Index Expanded (SCIE); (2) Social Sciences Citation Index (SSCI); (3) Arts & Humanities Citation Index (AHCI); (4) Emerging Sources Citation Index.

Publishers. For example, [Scopus](#) is owned by Elsevier and maintained by the [Scopus Content Selection and Advisory Board](#). Scopus includes journals in all disciplines, but the majority are science and technology journals.

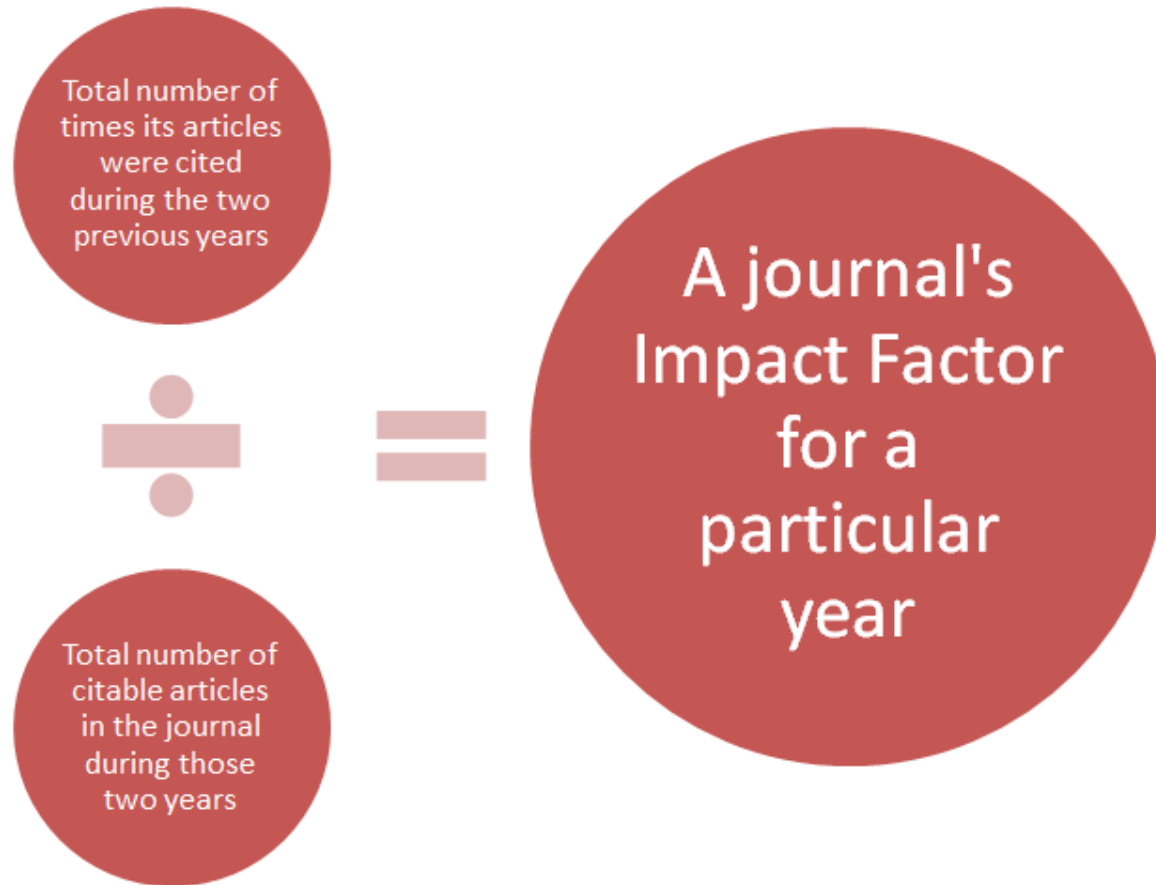
Impact Factor

- The impact factor (IF) or journal impact factor (JIF) of an academic journal is a [scientometric](#) index calculated by [Clarivate](#) that reflects the yearly mean number of [citations](#) of articles published in the last two years in a given journal, as indexed by Clarivate's [Web of Science](#).
- As a [journal-level metric](#), it is frequently used as a [proxy](#) for the relative importance of a journal within its field; journals with higher impact factor values are given the status of being more important, or carry more prestige in their respective fields, than those with lower values.

Measuring a Journal Impact Factor:

- Use [journal metrics](#) to understand the impact of a journal:
 - **CiteScore metrics** – helps to measure journal citation impact. Free, comprehensive, transparent and current metrics calculated using data from Scopus®, the largest abstract and citation database of peer-reviewed literature.
 - **SJR** – or SCImago Journal Rank, is based on the concept of a transfer of prestige between journals via their citation links.
 - **SNIP** – or Source Normalized Impact per Paper, is a sophisticated metric that accounts for field-specific differences in citation practices.
 - **JIF** – or Journal Impact Factor is calculated by Clarivate Analytics as the average of the sum of the citations received in a given year to a journal's previous two years of publications, divided by the sum of “citable” publications in the previous two years.
 - **H-index** – Although originally conceived as an author-level metric, the [H-index](#) has been being applied to higher-order aggregations of research publications, including journals.

Impact Factor



Example Calculation of an Impact Factor for 2009:

Cites in 2009 to articles published in:

- 2008 = 9000
- 2007 = 8000

Total recent citations in 2009 = 17000

Number of items published in:

- 2008 = 1500
- 2007 = 1200

Total number of recent items published = 2700

Impact Factor:

$$\frac{\text{Cites to recent item}}{\text{No. recent items}} = \frac{17000}{2700} = 6.296$$

Research Literature

Theory-based Literature

Philosophical Literature

Empirical or Practice-based Literature

Statistical Literature

Grey Literature

Research Literature:

Some edited books, journal articles, theses and government publications will employ research paradigms and methodologies to support a hypothesis.

Many of these publications are peer-reviewed, meaning they have been checked by a panel of experts before publication.

Most library databases offer a peer-reviewed checkbox which will filter search results in this way.

They will also allow you to filter results according to research methodology, focus group, geographical location and much more.

Research literature is an essential component of your literature review.

Theory-based Literature

Literature such as books, articles and reference sources will attempt to explain, describe, define and provide a background or theoretical framework for a field of inquiry.

These sources may include the original works of primary theorists as well as works which build upon, critique and discuss these primary sources while connecting it to the latest research.

This type of literature is also an important component of any literature review.

Philosophical Literature

Information sources such as books and articles, which deal with the underlying beliefs, attitudes and concepts that form the basic assumptions or building blocks within a profession or field of study.

This kind of literature formulates critical inquiry from either an ethical, epistemological, metaphysical or logical standpoint.

The extent to which you use philosophical literature will depend on the focus and subject matter of your review but it may be useful when constructing a background or theoretical base in your writing.

Empirical or Practice-based Literature

Encompasses literature that may be sourced via library databases and the internet.

Empirical literature is written by practitioners or experts in a particular field who wish to share their experiences and inform current practices in that area.

Literature types may include case studies, discussion papers, opinion pieces or reflective essays.

These resources may be useful to contextualise or support your literature review but should always be used in conjunction with research literature.

Statistical Reports

An important part of the research in any field of study, Statistical reporting and datasets are useful for description, analysis and comparison in your literature reviews.

When referring to statistical data in your reviews, ensure you source the most up-to-date research from reliable sources such as the Australian Bureau of Statistics, other government departments or authoritative institutions and organisations.

Grey Literature

Grey literature describes both unpublished or published documents, which are not controlled by commercial publishers.

These sources include: Government and non-government reports, policy statements and papers; conference proceedings and theses; news sources and bulletins; and technical reports and standards.

Again, depending on the focus of your assessment, this information may be used to supplement your primary resources, such as research and theoretical literature.

Grey Literature can be accessed by conducting a Google search or directly from relevant government, organisational and institutional websites.

Primary Literature

Secondary Literature

Tertiary Literature

Primary Literature

Primary sources means original studies, based on direct observation, use of statistical records, interviews, or experimental methods, of actual practices or the actual impact of practices or policies.

They are authored by researchers, contains original research data, and are usually published in a peer-reviewed journal.

Primary literature may also include conference papers, pre-prints, or preliminary reports.

Also called research.

Secondary Literature

Secondary literature consists of interpretations and evaluations that are derived from or refer to the primary source literature.

Examples include review articles (such as systematic reviews) and reference works.

Professionals within each discipline take the primary literature and synthesize, generalize, and integrate new research.

Tertiary Literature

Tertiary literature consists of a distillation and collection of primary and secondary sources such as textbooks, encyclopedia articles, and guidebooks or handbooks.

The purpose of tertiary literature is to provide an overview of key research findings and an introduction to principles and practices within the discipline.

Primary Literature Original research results in journals, dissertations, conference proceedings, correspondence	Secondary Literature Review articles, systematic reviews, meta-analysis, practice guidelines, monographs on a specific subject	Tertiary Literature Textbooks, encyclopedias, handbooks, newspapers
Sources: NEJM, JAMA	Sources: PubMed, CINAHL, Cochrane Library, Web of Science, Williams Obstetrics, Hurst's The Heart	Sources: Gale Encyclopedia of Genetic Disorders, Oxford Handbook of Internal Medicine